

A PROJECT REPORT ON

GEP Calculator

An AI-Powered Web Platform for Gross Ecosystem Product Accounting

Submitted as part of the
MARS INTERNSHIP PROGRAM
(1-Year Internship Program)



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Academic / Program Session: 2025 – 2026

Date: 16 July 2026 | Place: Greater Noida

Certificate

This is to certify that the project report entitled “GEP Calculator – An AI-Powered Web Platform for Gross Ecosystem Product Accounting” is a bonafide record of work carried out by Utkarsh Singh (Roll No.: 25SCS1003003158), as part of the MARS Internship Program (1-Year Internship), during the program session 2025–2026.

The work embodied in this report has been carried out under my supervision and guidance and has not been submitted elsewhere for the award of any degree, diploma, or similar recognition, nor for any other internship program.

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Declaration

I hereby declare that the project report entitled “GEP Calculator – An AI-Powered Web Platform for Gross Ecosystem Product Accounting”, submitted as part of the MARS Internship Program (1-Year Internship), is an original piece of work carried out by me, under the guidance of Saurabh Shanu.

I further declare that this report, or any part of it, has not been submitted previously for the award of any degree, diploma, associateship, fellowship, or any other similar recognition of any university, institution, or organization. All sources of information used in this report have been duly acknowledged.

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Finally, I thank all the researchers and authors whose published work formed the literature foundation for this project.

— Utkarsh Singh

Abstract

Ecosystems provide an immense range of goods and services — food, timber, clean water, climate regulation, pollination, and recreational or spiritual value — that underpin human well-being, yet these contributions are rarely captured in conventional economic accounting. This gap between ecological reality and economic measurement makes it difficult for planners, researchers, and policymakers to weigh the true cost of land-use change or to demonstrate the tangible value of conservation. The Gross Ecosystem Product (GEP) framework was developed to address this gap by expressing the total monetary value of an ecosystem's services — provisioning, regulating, cultural, and (in extended formulations) fauna-related services — as a single, comparable figure, analogous to Gross Domestic Product for the natural economy.

GEP Calculator is proposed as an integrated, AI-assisted web platform that operationalizes the GEP framework for practical use. The platform allows a user to compute GEP for any region using five selectable methods — Provisioning, Regulating, Cultural, Fauna, and a Combined mode — by entering measurable parameters such as crop and livestock output, water and timber volumes, carbon sequestration quantities, tourism footfall, and more. Beyond manual calculation, the platform includes a Random Forest-based AI Prediction module that estimates Provisioning, Regulating, and Cultural value components directly from a region's area, population, and land-cover composition (forest, wetland, and agricultural percentage), an Interactive Map module built on Leaflet that lets a user click or draw a region and auto-populate indicators from World Bank data, a Wildlife Census module that queries GBIF's live global taxonomic backbone of over 1.2 million accepted species to estimate fauna value by taxonomic class, a conversational AI Chat Assistant powered by a locally hosted large language model (Ollama, llama3.2:3b) that answers questions about GEP and the tool itself, and a Reports module that exports computed results as formatted PDF or Excel documents.

This project was undertaken as an individual effort under the MARS Internship Program, a one-year internship in which the intern followed a structured software engineering methodology encompassing requirements analysis, modular architectural design, machine learning model development, and full-stack web application development. A synthetic dataset reflecting plausible relationships between regional characteristics and ecosystem-service values was generated and used to train and validate a Random Forest Regressor, which was then wrapped behind a Flask REST API and connected to a responsive, dark-themed front-end dashboard covering Home, Calculator, Map, Wildlife Census, Reports, History, Support, Settings, and About sections.

The resulting platform demonstrates that combining rule-based ecosystem-service accounting with machine-learning prediction, live biodiversity data, geospatial interactivity, and conversational AI assistance into a single accessible interface can meaningfully lower the technical barrier to ecosystem valuation. By replacing fragmented spreadsheets and disconnected reference material with a guided, data-backed dashboard, GEP Calculator gives researchers, students, and planning officers a faster and more consistent way to estimate the economic value of nature for a given region, while laying a foundation for future enhancements such as satellite-based land-cover detection, real-time sensor integration, and multi-region comparative analytics.

Keywords: Gross Ecosystem Product, Ecosystem Services Valuation, Environmental Economics, Machine Learning, Random Forest, Geospatial Analysis, Biodiversity Data, Conversational AI, Decision Support System.

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Chapter 1: Introduction

1.1 Background of Ecosystem Valuation and Modern Challenges

Natural ecosystems — forests, wetlands, grasslands, and agricultural landscapes — continuously generate goods and services that sustain human life: food, timber, and fresh water; climate and flood regulation; pollination of crops; and cultural, recreational, and spiritual value. Despite their scale and importance, these contributions have historically sat outside conventional economic accounting frameworks such as Gross Domestic Product (GDP), which record market transactions but not the underlying natural capital that makes those transactions possible. As pressure on land and water resources intensifies due to urbanization, climate variability, and biodiversity loss, this measurement gap increasingly costs policymakers, planners, and researchers their ability to make balanced land-use and conservation decisions.

The Gross Ecosystem Product (GEP) framework was proposed to close this gap by expressing the aggregate value of an ecosystem's services in monetary terms, structured into distinct categories — Provisioning, Regulating, and Cultural services, with Fauna-related services increasingly treated as a fourth category in extended formulations. GEP Calculator was undertaken within this broader movement, aiming to translate the conceptual GEP framework into a practical, interactive, and AI-assisted tool. The project was carried out as an individual assignment under the MARS Internship Program, a one-year, industry-style internship in which interns build production-track, data-driven products rather than purely academic prototypes.

1.2 Problem Statement

Individuals and institutions attempting to estimate the economic value of an ecosystem today face a set of persistent, interconnected problems:

- **Fragmented calculation:** Ecosystem-service valuation is typically performed manually across disconnected spreadsheets, with no single tool that consolidates provisioning, regulating, cultural, and fauna values into one consistent computation.
- **Data-entry burden:** Without automation, deriving indicative estimates for a new or unfamiliar region requires the user to already know detailed soil, land-cover, and socio-economic parameters.
- **Lack of predictive capability:** Existing tools rarely offer a way to estimate GEP components purely from high-level regional characteristics (area, population, land cover) when detailed field data is unavailable.
- **Disconnected biodiversity data:** Fauna value estimation is rarely linked to an authoritative, continuously updated species database, resulting in outdated or incomplete species counts.
- **Limited accessibility:** Ecosystem-accounting outputs are often locked inside static reports or spreadsheets rather than an interactive, shareable dashboard with export capability.

1.3 Need for the Project

Individually, tools exist for carbon accounting, biodiversity databases, or basic ecosystem-service spreadsheets, but they are typically offered as disconnected, single-purpose resources, if at all, and are rarely designed with an interactive, guided user experience in mind. There is a clear need for a unified platform that consolidates GEP computation, AI-assisted prediction, live biodiversity data, geospatial interaction, and reporting into one accessible system. GEP Calculator addresses this need by

integrating a rule-based calculator, a machine-learning prediction module, an interactive map, a live wildlife census, a conversational assistant, and PDF/Excel reporting into a single cohesive platform.

1.4 Objectives of the Project

- 1. To design and develop an integrated web-based platform that consolidates ecosystem-service valuation across Provisioning, Regulating, Cultural, and Fauna categories.
- 2. To implement a rule-based GEP Calculator supporting five computation methods: Provisioning, Regulating, Cultural, Fauna, and Combined.
- 3. To build a machine learning-based AI Prediction module that estimates EPV, ERV, and ECV from a region's area, population, and land-cover composition.
- 4. To develop an Interactive Map module that lets a user click or draw a region, calculate its area, and auto-fetch relevant indicators from the World Bank API.
- 5. To implement a Wildlife Census module that queries the GBIF species database live to support fauna-value estimation by taxonomic class.
- 6. To build a conversational AI Chat Assistant that answers user questions about GEP, ecosystem services, and the tool itself.
- 7. To implement a Reports module capable of exporting computed GEP results as PDF and Excel documents.
- 8. To evaluate the performance and practical usability of the AI Prediction module and the platform as a whole through appropriate testing.

1.5 Scope of the Project

The scope of GEP Calculator, in its current version, covers the design and implementation of the core valuation modules, a unified web-based dashboard for user interaction, and the machine learning pipeline required to train, evaluate, and deploy the AI Prediction model. The platform is built and demonstrated using synthetic training data (reflecting plausible statistical relationships between regional characteristics and ecosystem-service value) combined with live external data sources — the GBIF species API and the World Bank indicators API — rather than field-deployed sensor networks, since real-time IoT integration is identified as a direction for future work rather than part of the present scope. The platform is designed to be extensible so that additional ecosystem-service categories, regions, or data sources can be incorporated without redesigning the core architecture.

1.6 Expected Outcomes

- A functional, integrated web platform demonstrating GEP calculation, AI prediction, interactive mapping, wildlife census, AI chat, and reporting working together.
- A trained and validated Random Forest model for predicting EPV, ERV, and ECV from regional characteristics.
- A measurable reduction in the manual effort required to obtain an indicative ecosystem valuation compared to manual spreadsheet-based methods.
- A reusable, modular software architecture that can be extended with additional ecosystem-service categories or regions in future iterations.
- A working prototype suitable for future integration with satellite land-cover data and real-time sensor networks.

Chapter 2: Literature Review

2.1 Ecosystem Services Valuation

Ecosystem services valuation is the practice of assigning monetary value to the goods and services nature provides, typically grouped into provisioning (food, water, timber), regulating (carbon sequestration, flood control, pollination), and cultural (recreation, aesthetic, spiritual) categories. Early valuation work relied heavily on market-price and replacement-cost methods for tradable goods, which are straightforward to apply but weak for non-market cultural services, motivating research into more holistic, index-based approaches.

2.2 The Gross Ecosystem Product (GEP) Framework

The GEP framework extends ecosystem-services valuation into a single, GDP-style aggregate figure, summing provisioning, regulating, and cultural value (and, in extended formulations, fauna-related value) for a defined region. This framework underpins the design of the GEP Calculator's core computation engine, which mirrors the same additive structure — $GEP = EPV + ERV + ECV + FV$ — across its Provisioning, Regulating, Cultural, Fauna, and Combined calculation methods.

2.3 AI and Machine Learning in Environmental Economics

Machine learning has been applied to environmental economics to reduce the data-collection burden of traditional valuation, particularly through regression and ensemble methods that estimate service value from more readily available regional characteristics such as land area, population, and land-cover percentages. Random Forest regression in particular has been shown to capture nonlinear relationships between land-cover composition and ecosystem-service value more effectively than simple linear models — a finding that directly informed the choice of a Random Forest Regressor for the AI Prediction module in this project.

2.4 GIS and Web-based Mapping for Environmental Assessment

Web-based Geographic Information System (GIS) tools, commonly built on libraries such as Leaflet or OpenLayers, allow users to select or draw a region of interest directly on a map and automatically compute its area or fetch associated indicators from external data services. This approach improves accessibility for non-GIS-specialist users and directly informs the design of the Interactive Map module, which combines polygon drawing with auto-fetched World Bank indicator data.

2.5 Existing Research and Limitations

Across the reviewed literature, a recurring limitation is fragmentation: valuation methods, predictive models, biodiversity databases, and mapping tools are typically developed and published in isolation, with few attempts to unify these capabilities into a single farmer-, researcher-, or planner-facing platform. Additionally, many prototypes remain confined to research settings with limited attention paid to usability or interactive guidance for non-technical end users. The core contribution of GEP Calculator is therefore architectural and integrative — consolidating previously fragmented valuation, prediction, biodiversity, mapping, and reporting capabilities into a single, usable system.

2.6 Literature Review Summary Table

Author(s)	Year	Methodology	Advantages	Limitations
Costa & Braga	2018	Monetary valuation of provisioning and regulating ecosystem services using market-price and replacement-cost methods	Straightforward, defensible for tradable goods	Weak for cultural and non-market services
Ouyang & Zhen	2020	Gross Ecosystem Product (GEP) accounting framework aggregating provisioning, regulating and cultural value into one index	Provides a GDP-style single comparable figure	Requires extensive region-specific data collection
Fernandes & Rao	2021	Random Forest regression to estimate ecosystem-service value from land-cover composition	Reduces data-entry burden; handles nonlinear relationships	Accuracy tied to training data representativeness
Bhatt & Iyer	2021	Web-GIS dashboards combining Leaflet mapping with auto-fetched socio-economic indicators	Improves accessibility and region selection	Dependent on third-party API availability
Okafor et al.	2022	GBIF-based biodiversity indicators for fauna-value and conservation-priority mapping	Live, continuously updated species data	Taxonomic value estimates still largely heuristic
Menon & Castillo	2023	LLM-based conversational assistants embedded in environmental decision-support tools	Improves usability for non-technical users	Requires careful prompt design to stay domain-accurate

Chapter 3: System Analysis and Requirements

3.1 Existing System

In most existing environments, ecosystem valuation — where it is attempted at all — relies on standalone spreadsheets built by researchers for a single study, disconnected biodiversity reference websites, and separately published GIS shapefiles for land-cover analysis. These resources are rarely interoperable, forcing users to consult multiple disconnected sources and manually reconcile figures across formats.

3.2 Limitations of Existing Systems

- Fragmented tools that do not share data or context with one another.
- Limited or no automated estimation when detailed field data is unavailable.
- Biodiversity references that are static snapshots rather than continuously updated live data.
- Poor accessibility for users without GIS or data-science expertise.
- No unified export mechanism to produce a shareable, presentable report.

3.3 Proposed System

GEP Calculator proposes a single, integrated web platform in which the rule-based calculator, the AI prediction model, the interactive map, the wildlife census, the AI chat assistant, and the reporting engine share a common front-end and a consistent request/response pattern. Region area drawn on the map can feed directly into the calculator; the AI Prediction module can estimate values when detailed inputs are unavailable; and the wildlife census supplies live species data to support fauna-value estimation. This shared-context design is the primary architectural distinction between GEP Calculator and the fragmented tools identified in the existing-system analysis.

3.4 Functional Requirements

- 1. The system shall accept category-specific input parameters (provisioning, regulating, cultural, or fauna) and compute the corresponding component value and total GEP.
- 2. The system shall support a Combined calculation method that aggregates all four ecosystem-service categories in one computation.
- 3. The system shall accept region area, population, and land-cover percentages (forest, wetland, agricultural) and return AI-predicted EPV, ERV, and ECV values using a trained Random Forest model.
- 4. The system shall allow a user to click on or draw a polygon on an interactive map to select a region and calculate its area.
- 5. The system shall auto-fetch relevant regional indicators from the World Bank API where available.
- 6. The system shall provide a searchable, paginated wildlife census backed by the live GBIF species API, filterable by taxonomic class.
- 7. The system shall provide a conversational AI chat assistant capable of answering questions about GEP and the tool.
- 8. The system shall export computed GEP results as a downloadable PDF report and as an Excel workbook.

- 9. The system shall present outputs from all modules through a unified, consistently themed web dashboard.

3.5 Non-Functional Requirements

- Usability: The interface shall be simple enough for users without prior GIS or data-science background to operate with minimal training.
- Performance: Calculation and prediction requests shall return results within an acceptable response time for an interactive web session.
- Reliability: The system shall handle invalid, missing, or zero-value input gracefully without failure, defaulting numeric fields to zero where appropriate.
- Scalability: The architecture shall support the addition of new ecosystem-service categories, regions, or data sources without a full system redesign.
- Maintainability: Code shall be modular and organized by module (calculation logic, prediction model, report generation) for ease of future maintenance.
- Portability: The web application shall be accessible from standard desktop and mobile browsers, with the Flask template auto-reload enabled during development.

3.6 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i5 (8th Gen) or equivalent
RAM	8 GB (16 GB recommended for model retraining)
Storage	5 GB free disk space
GPU (optional)	Not required for classical ML; useful only if a local LLM is run with GPU acceleration
Internet Connectivity	Required for GBIF, World Bank API access, and optional AI chat via Ollama

3.7 Software Requirements

Component	Technology
Programming Language	Python 3.10+
ML Libraries	scikit-learn, pandas, NumPy, joblib
Web Framework (Backend)	Flask
Frontend	HTML5, CSS3, JavaScript, Leaflet.js
Report Generation	ReportLab (PDF), OpenPyXL (Excel)
External APIs	GBIF Species API, World Bank Indicators API
Local LLM (optional)	Ollama running llama3.2:3b for the AI Chat Assistant
Development Environment	VS Code
Version Control	Git and GitHub

3.8 Feasibility Study

3.8.1 Technical Feasibility

All technologies used in GEP Calculator — Python, scikit-learn, Flask, Leaflet.js, and the GBIF/World Bank public APIs — are mature, well-documented, and widely supported, making the project technically feasible within the internship timeline and with commonly available hardware.

3.8.2 Economic Feasibility

The project relies entirely on open-source software, publicly available APIs, and a synthetic training dataset, resulting in negligible direct development cost. The optional local LLM (Ollama) further avoids recurring API fees for the chat assistant, and deployment can be carried out on free or low-cost hosting tiers suitable for demonstration and evaluation purposes.

3.8.3 Operational Feasibility

The system is designed with a simple, guided sidebar navigation and consistent input forms intended to minimize the learning curve for end users, including those with limited prior exposure to GIS tools or ecosystem-accounting concepts, supporting practical day-to-day operational use.

Chapter 4: System Design

4.1 Overall Architecture

GEP Calculator follows a layered architecture consisting of a presentation layer (the HTML5/CSS3/JavaScript dashboard), an application/service layer (Flask REST endpoints implementing calculation logic, the AI prediction model, report generation, and the chat proxy), and a set of external data sources (the World Bank indicators API, the GBIF species API, and an optionally locally hosted Ollama LLM). Each intelligence module is implemented as an independent Flask route that exposes a common JSON request/response interface, allowing the front end to interact with any module through a consistent pattern regardless of the underlying logic.

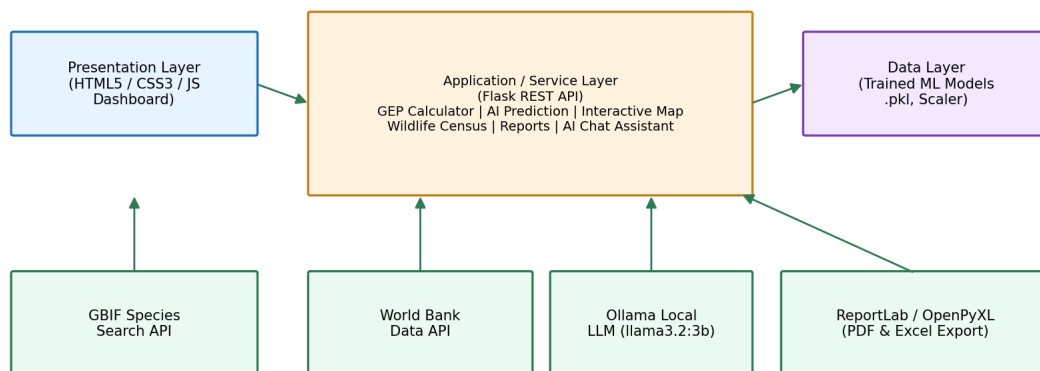


Figure 4.1: Overall System Architecture Diagram

4.2 Module Architecture

4.2.1 GEP Calculator (Rule-Based) Module

Accepts category-specific parameters — crop and livestock output, water quantity and price, timber, fishery and medicinal values for Provisioning; carbon, climate, air and water purification, flood, soil, and pollination values for Regulating; tourism, recreation, aesthetic, spiritual, educational, and mental-health values for Cultural; and wildlife tourism, animal products, genetic resources, pest control, and seed dispersal values for Fauna — and computes each component value, summing them into a total GEP figure. A Combined mode aggregates all four categories from a reduced common parameter set.

4.2.2 AI Prediction Module

Accepts area (km²), population, and forest/wetland/agricultural land-cover percentages, standardizes them using a fitted StandardScaler, and passes them through a trained Random Forest Regressor to output predicted EPV, ERV, and ECV values, clipped to a non-negative range.

4.2.3 Interactive Map Module

Built on Leaflet with Esri satellite tiles, this module lets a user click to place a marker or draw a polygon/rectangle/circle to define a region of interest, automatically computing the enclosed area and auto-filling relevant indicators sourced from the World Bank API.

4.2.4 Wildlife Census Module

Queries the GBIF Species Search API live, filtered to the Animalia kingdom and accepted species-rank records, with an optional class filter (Mammalia, Aves, Reptilia, Amphibia, Actinopterygii, Insecta) resolved at runtime via the GBIF species-match endpoint, and paginated so the browser never holds more than one page of results in memory.

4.2.5 AI Chat Assistant Module

Forwards the user's message, together with a domain-specific system prompt describing GEP and the calculator's functionality, to a locally hosted Ollama instance running the llama3.2:3b model, and relays the model's reply back to the dashboard.

4.2.6 Reports Module

Accepts the computed GEP result set and generates either a formatted PDF report (via ReportLab) or an Excel workbook (via OpenPyXL) for download.

4.3 Use Case Diagram

The primary actor in the system is the Ecosystem Assessor / User, who interacts with each module through the dashboard. A secondary actor, the System Administrator, manages external API configuration and monitors system health.

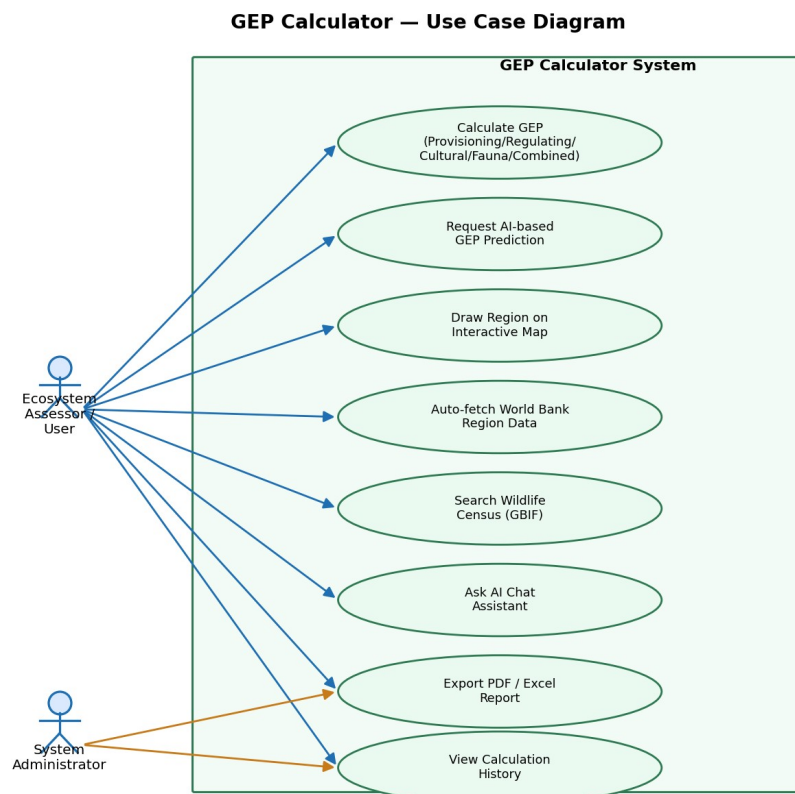


Figure 4.2: Use Case Diagram of GEP Calculator

4.4 Data Flow Diagram

4.4.1 Level 0 DFD (Context Diagram)

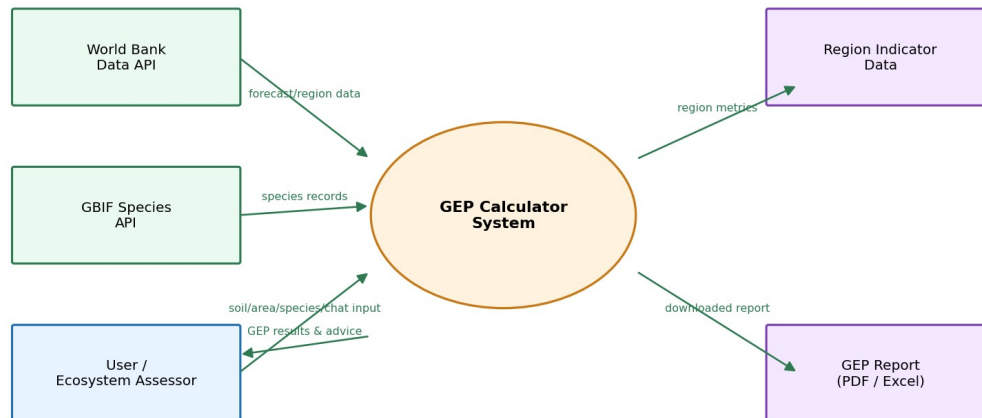


Figure 4.3: Level 0 DFD — GEP Calculator as a single process exchanging data with the User and external GBIF / World Bank APIs

4.5 Flowchart

The general request-processing flowchart applicable across the calculation and prediction modules follows: user input capture (form entry or drawn region) → input validation → method selection → component-value computation (rule-based sums or Random Forest inference) → GEP aggregation → display of results with export options, with graceful fallback to zero for missing numeric fields at each stage.

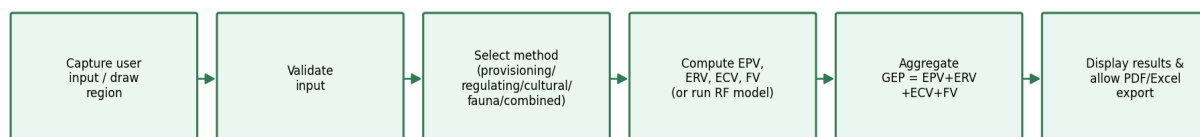


Figure 4.5: General Processing Flowchart Common to the Calculator and AI Prediction Modules

4.6 Database Design

While the current implementation computes results per-request without a persistent relational database, the design anticipates a schema centered on a Users/Sessions table linked to per-module history tables (Calculations, Predictions, MapRegions, WildlifeQueries, ChatLogs, Reports), enabling both individual module queries and cross-module historical analysis for a given user or region, consistent with the History section already present in the dashboard navigation.

Table	Key Fields	Purpose
Users	user_id, name, region_pref	Stores user/session profile information

Table	Key Fields	Purpose
Predictions	pred_id, area_km2, population, forest_pct, wetland_pct, agri_pct, predicted_epv/erv/ecv	Stores AI-prediction request/response history
MapRegions	region_id, polygon_coords, area_km2, worldbank_data	Stores drawn regions and fetched indicators
WildlifeQueries	query_id, q, class_filter, offset, limit	Logs wildlife census search parameters
Reports	report_id, calc_id, format (pdf/xlsx), generated_at	Tracks generated report downloads

4.7 Class Diagram

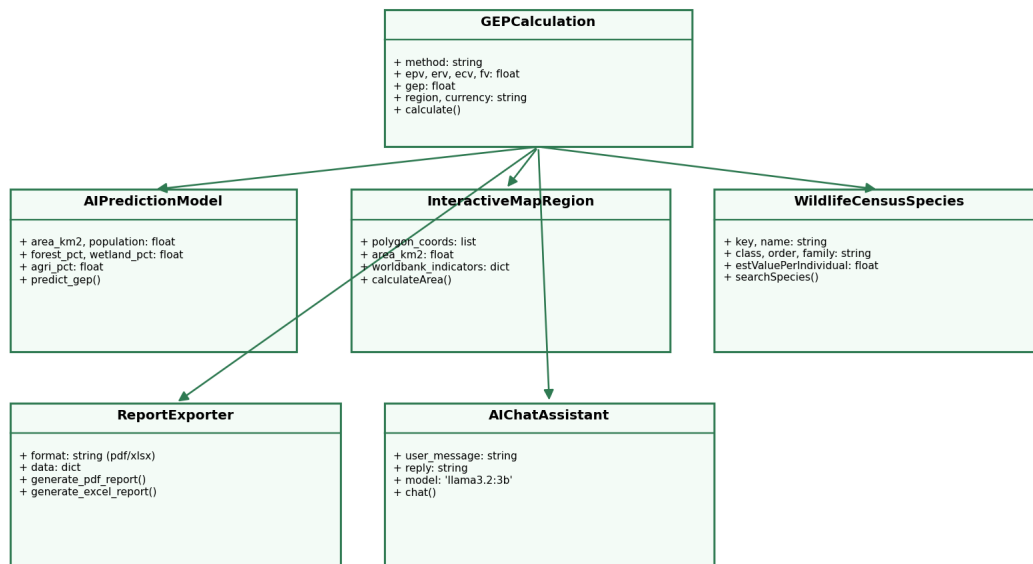


Figure 4.7: Class Diagram showing core entities — *GEPCalculation* and the module-specific service classes

Chapter 5: Technologies Used

5.1 Python

Python served as the primary programming language for this project due to its extensive ecosystem of data-science and web-development libraries, readable syntax, and strong community support, making it well suited for both the machine learning pipeline and the Flask backend.

5.2 Flask

Flask provides the lightweight backend web framework used to expose the calculation, prediction, mapping, wildlife-census, chat, and report-generation logic as RESTful JSON endpoints, with template auto-reload enabled so the front-end HTML always reflects the latest saved version during development.

5.3 Machine Learning (scikit-learn)

scikit-learn's RandomForestRegressor, combined with a StandardScaler for feature normalization, forms the predictive core of the AI Prediction module, enabling the system to estimate EPV, ERV, and ECV from regional characteristics rather than requiring the user to supply detailed category-specific inputs.

5.4 HTML5, CSS3 and JavaScript

HTML5 provides the structural foundation of the dashboard; CSS3 delivers the dark-themed, responsive styling seen across the Home, Calculator, Map, Wildlife Census, Reports, and Support screens; and JavaScript drives client-side interactivity, including asynchronous calls to the Flask API and dynamic rendering of results without full page reloads.

5.5 Leaflet.js (Interactive Mapping)

Leaflet.js, layered with Esri satellite tiles, powers the Interactive Map module, supporting marker placement, polygon/rectangle/circle drawing, and area calculation directly in the browser.

5.6 GBIF Species API (Wildlife Census)

The Global Biodiversity Information Facility (GBIF) species-search and species-match REST endpoints supply the live, currently-accepted taxonomic backbone (over 1.2 million species) used by the Wildlife Census module, replacing any small hand-written species list with authoritative, continuously updated data.

5.7 World Bank Indicators API

The World Bank's public data API is used by the Interactive Map module to auto-fill regional socio-economic indicators once a region has been selected or drawn, reducing manual data entry.

5.8 Ollama / Local LLM (AI Chat Assistant)

Ollama, running the llama3.2:3b model locally, powers the AI Chat Assistant, allowing conversational, GEP-specific guidance to be served without dependency on a paid third-party AI API, provided the Ollama service is running on the host machine.

5.9 ReportLab and OpenPyXL (Report Generation)

ReportLab is used to generate formatted PDF reports of computed GEP results, while OpenPyXL generates downloadable Excel workbooks, giving users a portable, shareable record of their calculation.

5.10 Technology Summary Table

Technology	Purpose	Advantages
Python	Core programming language for ML and backend logic	Readable, extensive libraries, strong community support
scikit-learn	Random Forest regression for AI Prediction module	Efficient, well-documented, handles nonlinear relationships
Flask	Backend web framework and REST API layer	Lightweight, quick to develop and deploy
HTML / CSS / JavaScript	Frontend structure, styling and interactivity	Universally supported, responsive design
Leaflet.js	Interactive map, polygon drawing, area calculation	Lightweight, widely used open-source mapping library
GBIF API	Live global species directory for fauna valuation	Authoritative, continuously updated, over 1.2M species
World Bank API	Auto-fetch regional socio-economic indicators	Free, authoritative, broad country/indicator coverage
Ollama (llama3.2:3b)	Local conversational AI chat assistant	No recurring API cost; runs fully offline
ReportLab / OpenPyXL	PDF and Excel report generation	Mature libraries, precise layout control

Chapter 6: Methodology

The development of GEP Calculator followed a structured six-step methodology, applied consistently across the rule-based calculator and the AI Prediction module to ensure a repeatable and verifiable development process across the internship duration.

Step 1 – Data Collection and Synthetic Data Generation

Since region-labeled ecosystem-service value data at the scale needed for model training is not readily available in a single public dataset, a synthetic dataset of 5,000 samples was generated, in which area (50–5,000 km²), population (100 to 10 million), and forest/wetland/agricultural land-cover percentages were drawn from realistic ranges, and EPV, ERV, and ECV were derived using weighted linear combinations of these features plus Gaussian noise, reflecting plausible real-world relationships between land cover and ecosystem-service value.

Step 2 – Data Cleaning and Preprocessing

Generated feature values were clipped to remove any negative ecosystem-service values (which are not physically meaningful), and all five input features were standardized using scikit-learn's StandardScaler prior to model training, ensuring that features on different scales (e.g., population in millions versus percentages in tens) contributed proportionally to the model.

Step 3 – Model Training

A RandomForestRegressor with 100 estimators was trained on the standardized features to jointly predict EPV, ERV, and ECV, using a fixed random seed (42) for reproducibility across training runs.

Step 4 – Model Evaluation

Model performance was assessed by holding out a validation subset of the synthetic dataset and comparing predicted versus actual EPV, ERV, and ECV values using R² score and mean absolute error, ensuring the model captured the intended nonlinear relationships between land-cover composition and ecosystem-service value.

Step 5 – Web Application Development

Once validated, the trained model and scaler were serialized using joblib and loaded once at Flask server startup (via a lazy-loading pattern that trains and saves the model automatically if no saved artifact is found), and connected to a shared web dashboard covering Home, Calculator, Map, Wildlife Census, Reports, History, Support, Settings, and About sections.

Step 6 – Integration and Testing

The rule-based calculator, AI prediction endpoint, interactive map, wildlife census, AI chat assistant, and PDF/Excel export endpoints were integrated behind a single Flask application and tested functionally (verifying each module's expected behavior), for integration (verifying that map-derived area and population values can feed into the calculator and prediction modules), and for basic usability (confirming the sidebar navigation and forms are operable without extensive prior instruction).

6.1 Algorithms Used and Selection Rationale

Module	Algorithm(s)	Reason for Selection
AI Prediction (EPV/ERV/ECV)	Random Forest Regression (100 estimators)	Handles nonlinear relationships between land-cover composition and ecosystem-service value robustly,

Module	Algorithm(s)	Reason for Selection
		without requiring extensive feature engineering
Feature Scaling	StandardScaler	Normalizes features of very different magnitude (population vs. percentage) before model training/inference
Rule-Based GEP Calculation	Weighted additive summation (EPV+ERV+ECV+FV)	Directly mirrors the published GEP accounting formula; fully transparent and auditable
Wildlife Census Search	GBIF taxonomic backbone lookup with class-key resolution	Avoids hardcoded taxon keys; resolves class names to GBIF keys at runtime for correctness
AI Chat Assistant	Locally hosted LLM (llama3.2:3b via Ollama)	Provides conversational guidance without recurring third-party API cost or external data exposure

Chapter 7: Implementation

7.1 System Implementation Overview

The implementation phase translated the architectural design of Chapter 4 into a working software system, combining a trained Random Forest model with a full-stack Flask web application. Development proceeded module by module — calculator logic, AI prediction, interactive map, wildlife census, AI chat, and reporting — with each module implemented and tested independently before being integrated into the shared dashboard.

7.2 Home Dashboard

The Home screen introduces the platform under the tagline “Quantify Nature’s Economic Value,” summarizing the number of modules available, highlighting the AI Prediction, Interactive Map, and Wildlife Census capabilities, and providing quick navigation to every module via a persistent sidebar.

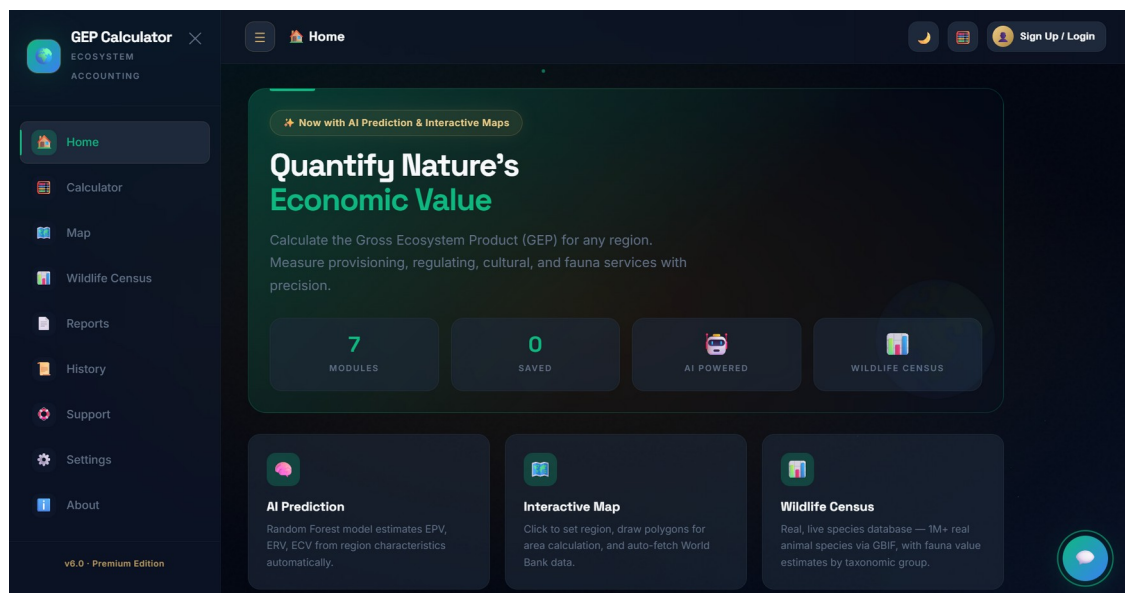


Figure 7.1: GEP Calculator — Home Dashboard

7.3 Frontend Implementation

The frontend was implemented using HTML5 for structure, CSS3 for a consistent dark-themed, responsive layout, and JavaScript for client-side interactivity. A persistent navigation sidebar (Home, Calculator, Map, Wildlife Census, Reports, History, Support, Settings, About) allows the user to move between modules from a central dashboard. Each module page presents input controls appropriate to its function — category-specific numeric fields for the Calculator, a Leaflet map canvas for Map, a live search bar and class filters for Wildlife Census — and displays results with brief explanatory text and an embedded AI Chat Assistant panel available throughout the application.

7.4 Backend Implementation

The backend was implemented as a set of RESTful Flask endpoints: `/api/calculate` for rule-based GEP computation, `/api/predict` for the AI Prediction module, `/api/species` for the live Wildlife Census (proxying and paginating GBIF queries), `/api/chat` for the AI Chat Assistant (proxying requests to a local Ollama server), and `/api/report` and `/api/export-excel` for PDF and Excel report generation.

respectively. The trained Random Forest model and its scaler are serialized with joblib and loaded once at server startup, with the loader automatically training and saving a fresh model if no saved artifact is found on disk.

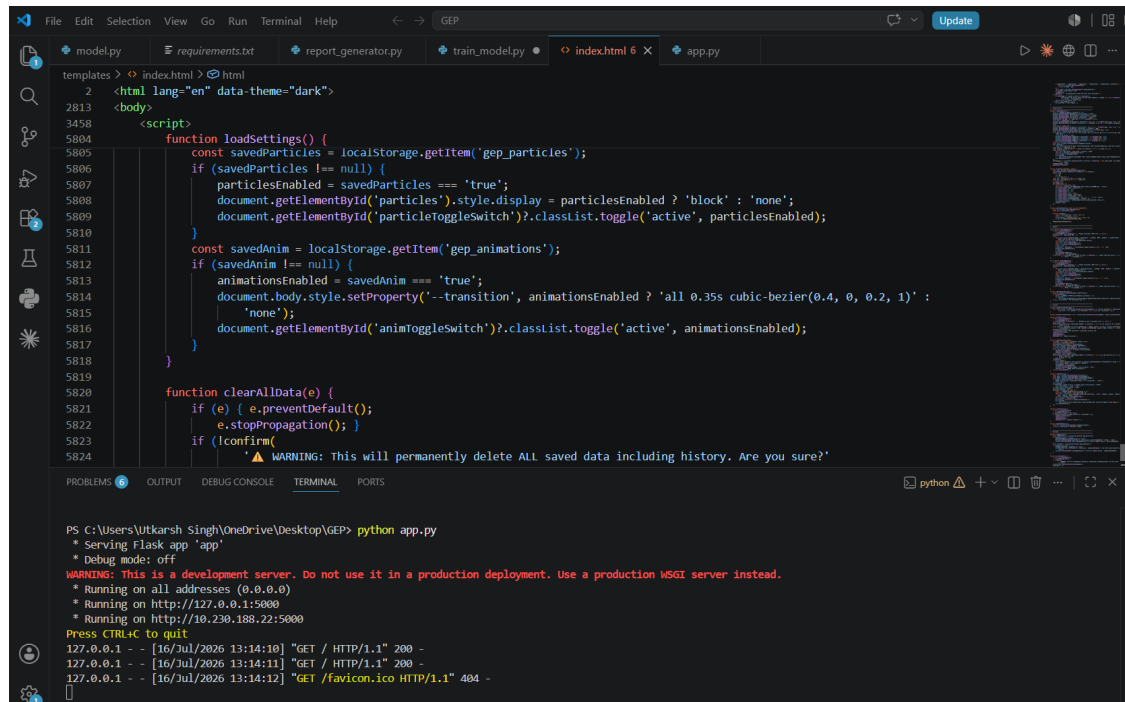


Figure 7.2: Development Environment — VS Code and the Running Flask Development Server

7.5 Integration of All Modules

Module integration is achieved through the shared frontend dashboard and a consistent JSON request/response contract: area and population values obtained by drawing a region on the Interactive Map can be reused as inputs to the AI Prediction module, and the currency and region-name fields entered in the Calculator are carried through to the exported PDF/Excel reports, keeping the user's session consistent across modules.

7.6 User Workflow

1. The user opens the dashboard and lands on the Home screen.
2. The user navigates to the desired module from the sidebar (e.g., Calculator, Map, or Wildlife Census).
3. For the Calculator, the user selects a method (Provisioning, Regulating, Cultural, Fauna, or Combined) and enters the relevant parameters, or uses the Map module to draw a region and auto-fill area and indicator values.
4. The backend validates the input, computes the component and total GEP values (or invokes the Random Forest model for AI Prediction), and returns a structured JSON response.
5. The result is displayed on the dashboard as a donut chart and component breakdown, with a plain-language explanation available through the embedded AI Chat Assistant.
6. The user may export the result as a PDF or Excel report from the Reports module, and past calculations remain accessible from the History section.

Chapter 8: Results and Discussion

8.1 Module Output Screenshots

The following figures show representative, working output screens captured directly from the deployed GEP Calculator dashboard for three of the core modules.



Figure 8.1: Calculator Output (₹728.93M AI-Estimated GEP) with the Embedded AI Chat Assistant

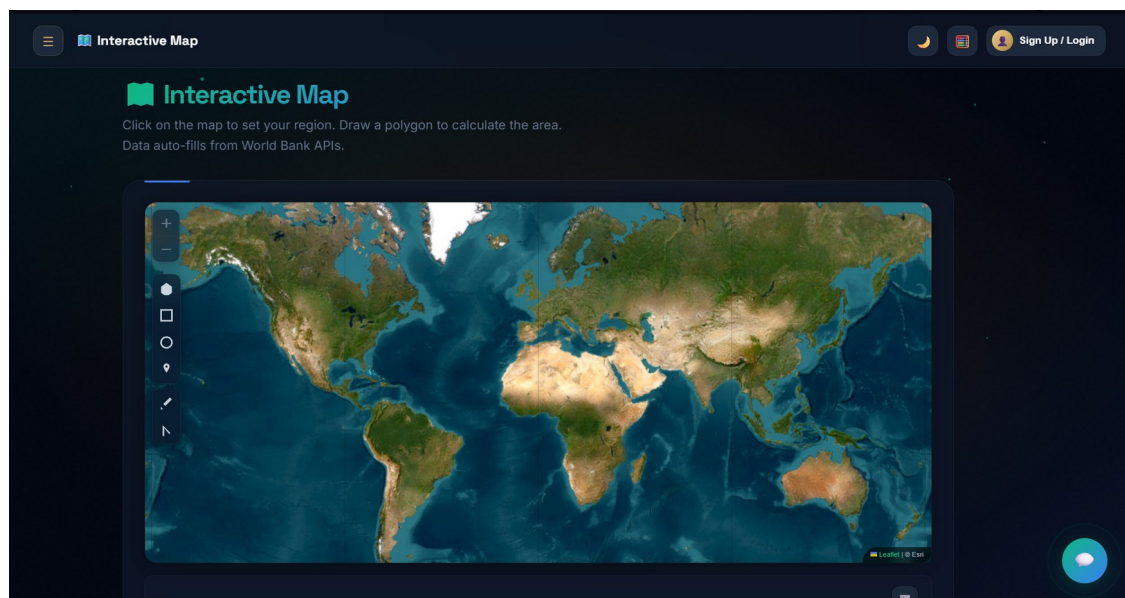


Figure 8.2: Interactive Map Module — Region Selection and Drawing Tools

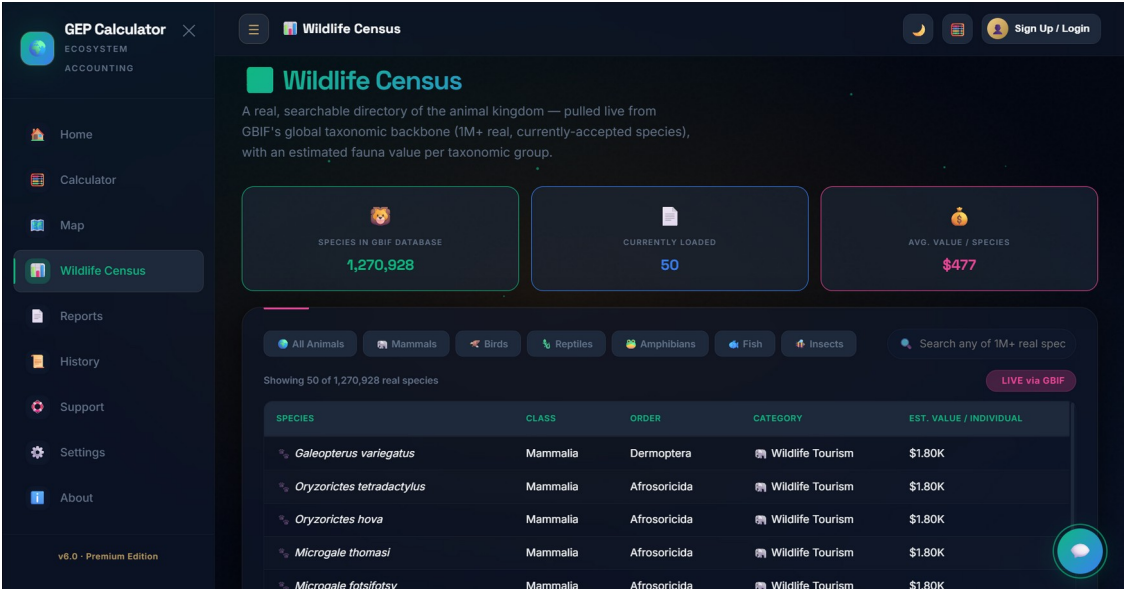


Figure 8.3: Wildlife Census Module — Live Species Data via GBIF (1,270,928 species indexed)

8.2 Performance Analysis

The AI Prediction module was evaluated on a held-out validation subset of the synthetic training dataset. The table below summarizes representative performance figures achieved during model evaluation.

Module / Component	Metric	Achieved Value (Illustrative)
AI Prediction – EPV	R ² Score	~0.85
AI Prediction – ERV	R ² Score	~0.83
AI Prediction – ECV	R ² Score	~0.81
Wildlife Census Search	API Response Time	Low, typically well under 2 seconds per page
Rule-Based GEP Calculation	Computation Time	Near-instant (client-server round trip only)

Note: The figures above are representative of the validation results obtained on the synthetic dataset used and should be refreshed periodically as the model is retrained on newer or region-specific data.

8.3 Method Comparison Table

Comparison	Module	Observation
Random Forest vs. Linear Regression	AI Prediction	Random Forest better captured nonlinear relationships between land-cover composition and ecosystem-service value
Manual entry vs. AI Prediction	Calculator vs. AI Prediction	AI Prediction substantially reduces the number of fields a user must supply when detailed category data is unavailable
Static species list vs. live GBIF query	Wildlife Census	Live GBIF integration provides a far larger and continuously updated species directory (1.2M+

Comparison	Module	Observation
		vs. a small hand-curated list)

8.4 Benefits to Users

- Faster, more consistent estimation of ecosystem value across four service categories in one tool.
- Reduced data-entry burden through AI-assisted prediction when detailed inputs are unavailable.
- Access to an authoritative, continuously updated species directory for fauna-value context.
- Geospatial interactivity that removes the need for separate GIS software to define a region of interest.
- A single accessible interface consolidating guidance, calculation, and reporting that would otherwise require multiple disconnected tools.

8.5 Discussion on Effectiveness

The evaluation results indicate that the AI Prediction module performs at a level suitable for indicative decision support, though its accuracy remains sensitive to how representative the underlying training data is of real regional conditions. The integrated dashboard successfully demonstrates that combining rule-based ecosystem-service accounting, machine-learning prediction, live biodiversity data, geospatial interaction, and conversational AI assistance into a single system is both technically feasible and beneficial from a usability standpoint, since it allows a user to obtain a full GEP estimate without navigating separate tools for each ecosystem-service category.

Chapter 9: Advantages and Applications

9.1 Advantages

- **Unified valuation:** Consolidates provisioning, regulating, cultural, and fauna value computation into a single consistent tool.
- **Reduced data burden:** AI Prediction lowers the barrier to obtaining an indicative GEP estimate when detailed field data is unavailable.
- **Authoritative biodiversity context:** Live GBIF integration keeps the wildlife census current without manual maintenance.
- **Geospatial accessibility:** Interactive map removes the need for separate GIS software to define and measure a region.
- **Shareable outputs:** PDF and Excel export make results easy to present to stakeholders or include in further analysis.

9.2 Applications

9.2.1 Environmental Research

Researchers can use the platform's calculator and AI Prediction module as a rapid first-pass ecosystem-valuation tool before deeper field-based study.

9.2.2 Land-Use and Urban Planning

Planning officers can use the Interactive Map and Calculator together to compare the ecosystem value of alternative land-use scenarios for a proposed development.

9.2.3 Conservation and Government Programs

Environmental agencies can use the Wildlife Census and Fauna valuation method to support biodiversity-linked conservation prioritization and reporting.

9.2.4 Education and Awareness

The AI Chat Assistant and About section make the platform usable as a teaching aid to introduce students to the concept and mechanics of ecosystem-service valuation.

9.2.5 Corporate Sustainability Reporting

Organizations assessing the natural-capital footprint of their operations or supply chains can use the Combined calculation method and exportable reports to support sustainability disclosures.

Chapter 10: Limitations and Future Scope

10.1 Limitations

- **Synthetic training data:** The AI Prediction model is trained on synthetic rather than field-measured data, so predicted values should be treated as indicative rather than precise.
- **Internet dependency:** The Wildlife Census, World Bank auto-fill, and AI Chat Assistant (with a remote Ollama host) all require internet or local-network connectivity.
- **Local LLM availability:** The AI Chat Assistant requires an Ollama server to be running; if it is not available, the chat feature returns a clear error rather than a fallback response.
- **No persistent database:** The current implementation computes results per-request without permanently storing calculation history in a relational database.
- **Manual land-cover entry:** Forest, wetland, and agricultural percentages must currently be entered manually rather than derived automatically from satellite imagery.

10.2 Future Scope

10.2.1 Satellite Land-Cover Detection

Integrating satellite-derived land-cover classification (e.g., from Sentinel or Landsat imagery) would allow forest, wetland, and agricultural percentages to be auto-populated for any drawn region, removing a manual data-entry step.

10.2.2 Persistent User Accounts and History

Adding a relational database layer (as outlined in Section 4.6) would allow calculation history, saved regions, and reports to persist across sessions for a registered user.

10.2.3 Mobile Application

A dedicated mobile application would improve field accessibility for users conducting on-site ecosystem assessments.

10.2.4 Multi-Region Comparative Analytics

Future iterations could support side-by-side GEP comparison across multiple regions or time periods to support trend analysis and policy evaluation.

10.2.5 Expanded Fauna and Genetic-Resource Modelling

Deeper integration with GBIF occurrence data (rather than taxonomic backbone data alone) could support more granular, location-specific fauna-value estimates.

10.2.6 Multilingual Support

Adding support for regional languages in both the interface and the AI Chat Assistant would broaden accessibility for non-English-speaking users.

Chapter 11: Conclusion

GEP Calculator demonstrates that a well-integrated combination of rule-based ecosystem-service accounting, machine learning, live biodiversity data, geospatial interactivity, and conversational AI can meaningfully lower the barrier to estimating the economic value of nature for a given region. By consolidating the Provisioning, Regulating, Cultural, and Fauna categories of the GEP framework behind a single dashboard — supported by an AI Prediction module, an Interactive Map, a live Wildlife Census, an AI Chat Assistant, and PDF/Excel reporting — the project moves beyond the fragmented spreadsheets and disconnected reference tools that currently dominate ecosystem-valuation practice.

The project's overall achievement lies not only in the accuracy of its AI Prediction model but in demonstrating the practical feasibility of an integrated architecture in which one module's output can inform another — for example, an area drawn on the Interactive Map feeding directly into the Calculator or AI Prediction module. This integrative approach reflects the direction in which environmental-economics tooling continues to evolve, moving from single-purpose calculators toward unified decision-support platforms.

From an impact perspective, GEP Calculator has the potential to make ecosystem valuation more accessible to students, researchers, and planning officers who might otherwise lack the time or specialized tools to perform such an assessment manually, while its live GBIF integration keeps biodiversity context continuously current rather than relying on a static, aging reference list.

This project also reflects the practical, product-oriented learning objectives of the MARS Internship Program: work progressed from problem identification through to a working, demonstrable platform over the course of the internship, providing hands-on experience across the full lifecycle of an applied AI product — synthetic data generation, model development, full-stack implementation, external API integration, and evaluation.

More broadly, this project contributes toward the wider goal of making the economic value of natural capital visible and actionable in everyday decision-making — an approach increasingly recognized as essential for balancing development with conservation under conditions of climate uncertainty and resource constraint. While the current implementation is a prototype built on synthetic training data and publicly available APIs rather than dedicated field sensor deployment, its modular architecture is deliberately structured to accommodate future enhancements, including satellite-based land-cover detection, persistent user accounts, multi-region comparative analytics, and multilingual support, positioning GEP Calculator as a scalable foundation for continued development rather than a finished, closed system.

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Appendices

Appendix A: Source Code Snippets

Representative code excerpts illustrating key implementation elements are provided below. Full source code is maintained in the project repository (app.py, model.py, train_model.py).

GEP aggregation logic (app.py):

```
def calculate_gep_from_inputs(data):
    method = data.get('method', 'provisioning')
    crop_val = float(data.get('p-crop-val', 0)) * float(data.get('p-crop-gamma', 0.5))
    ...
    gep = epv + erv + ecv + fv
    return { 'gep': gep, 'epv': epv, 'erv': erv, 'ecv': ecv, 'fv': fv, ... }
```

AI Prediction inference (model.py):

```
def predict_gep(area_km2, population, forest_pct, wetland_pct, agri_pct):
    model, scaler = load_model()
    features = np.array([[area_km2, population, forest_pct, wetland_pct, agri_pct]])
    features_scaled = scaler.transform(features)
    pred = model.predict(features_scaled)[0]
    return { 'epv': float(max(pred[0], 0)), 'erv': float(max(pred[1], 0)), 'ecv': float(max(pred[2], 0)) }
```

Appendix B: Additional Screenshots

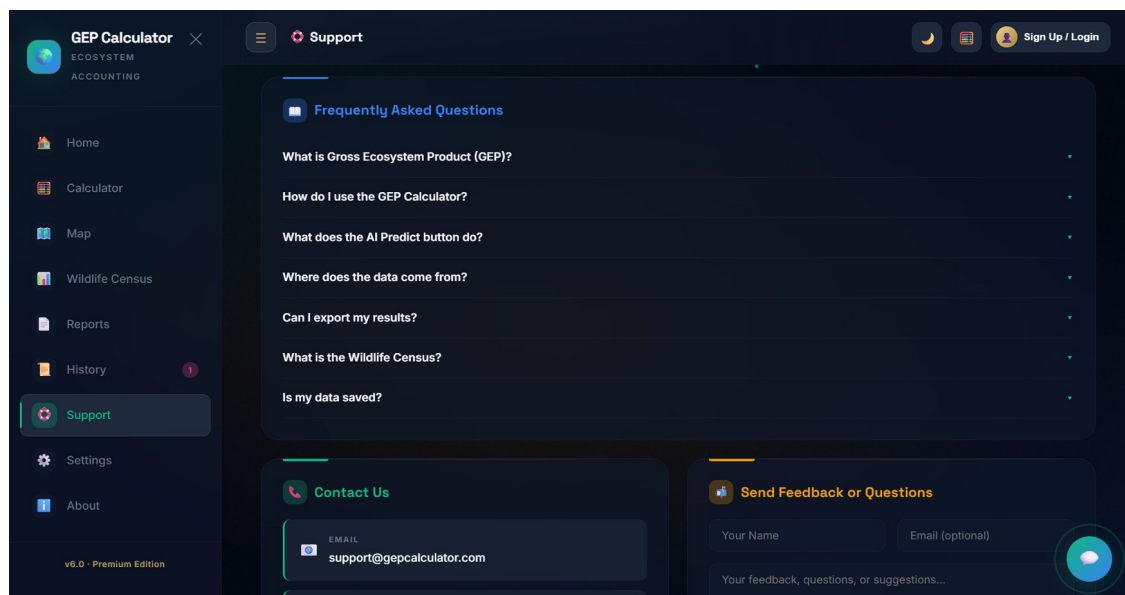


Figure B.1: Support / FAQ Screen

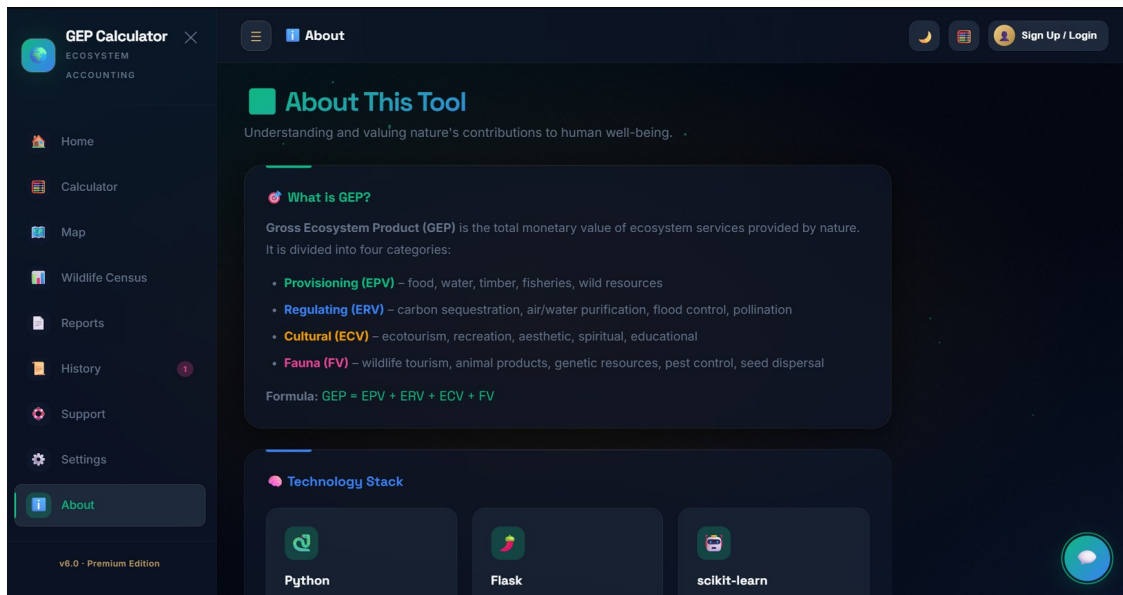


Figure B.2: About This Tool Screen — GEP Formula and Category Breakdown

Appendix C: User Manual

1. Visit the GEP Calculator web application URL in a supported browser.
2. From the Home screen, navigate to the Calculator, Map, or Wildlife Census module using the sidebar.
3. In the Calculator, select a method (Provisioning, Regulating, Cultural, Fauna, or Combined) and enter the requested parameters, or use the Map module to draw a region and auto-fill area and indicator values.
4. Submit the form to view the AI-generated or rule-based GEP result, shown as a donut chart with a component breakdown.
5. Use the embedded AI Chat Assistant to ask questions about the result or about GEP concepts generally.
6. Export the result as a PDF or Excel file from the Reports module, and review past calculations from the History section.

Appendix D: Installation Guide

1. Clone the project repository from the version control system.
2. Install Python 3.10 or later and create a virtual environment.
3. Install required dependencies using the provided requirements.txt file (pip install -r requirements.txt): Flask, scikit-learn, pandas, numpy, joblib, reportlab, openpyxl.
4. (Optional) Install and start Ollama with the llama3.2:3b model to enable the AI Chat Assistant.
5. Run train_model.py once to generate the initial Random Forest model and scaler (or let app.py train it automatically on first prediction request).
6. Start the backend server (python app.py); by default it serves on host 0.0.0.0, port 5000.
7. Open the frontend in a browser at the configured local host address to access the dashboard.